

Linux Kernel 6.5 Officially Released, This Is What's New

Marius Nestor

2023-08-27T22:05:03

After seven weeks of RCs, Linux kernel 6.5 is here with new features like Wi-Fi 7 support, MIDI 2.0 support in ALSA, ACPI support for the RISC-V architecture, Landlock support for UML (User-Mode Linux), better support for AMD “Zen” systems, as well as user-space support for the ARMv8.8 memcpy/memset instructions. Linux 6.5 includes hardware support for Lenovo Yoga Book yb1-x90f/l and Nextbook Ares 8A tablets, Dell Studio 1569 (ACPI backlight quirks), Lenovo ThinkPad X131e (3371 AMD version), and Apple iMac11,3 laptops. Of course, there are also numerous new and updated drivers for various devices to provide users with top-notch hardware support. Linux kernel 6.5 is available for download right now from Linus Torvalds’ Git tree or from the kernel.org website. If that’s not your cup of tea, you’ll have to wait for Linux 6.5 to arrive in the stable software repositories of your GNU/Linux distribution to update.

Just a couple of days after celebrating its 32nd anniversary, Linus Torvalds announced today the final release of the Linux 6.5 kernel series as a major update that introduces several new features, updated and new drivers for better hardware support, and other changes.

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Also new in Linux 6.5 is Intel TPMI (Topology Aware Register and PM Capsule Interface) support for the power capping subsystem and a TPMI interface driver for Intel RAPL, and the “runnable boosting” feature in the EAS balancer to improve CPU utilization for specific workloads.

This release also improves SMP scheduling’s load balancer to recognize SMT cores with more than one busy sibling and allows lower-priority CPUs to pull tasks to avoid superfluous migrations, and improves EXT4 file system’s journalling, block allocator subsystems, and performance for parallel DIO overwrites.

The performance for the Btrfs file system has been improved as well in this release, which also introduces SGX/HPE Ultraviolet support for the Sub-NUMA clustering (SNC).

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Of course, there are also numerous new and updated drivers for various devices to provide users with top-notch hardware support. Among these, there are drivers for the AHT20 temperature and humidity sensor, HP WMI sensors, and the ROG Crosshair X670E Hero motherboard.

Linux kernel 6.5 is available for download right now from Linus Torvalds' Git tree or from the kernel.org website. However, this is the source code, which needs to be compiled manually. If that's not your cup of tea, you'll have to wait for Linux 6.5 to arrive in the stable software repositories of your GNU/Linux distribution to update.

Last updated 2 months ago

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